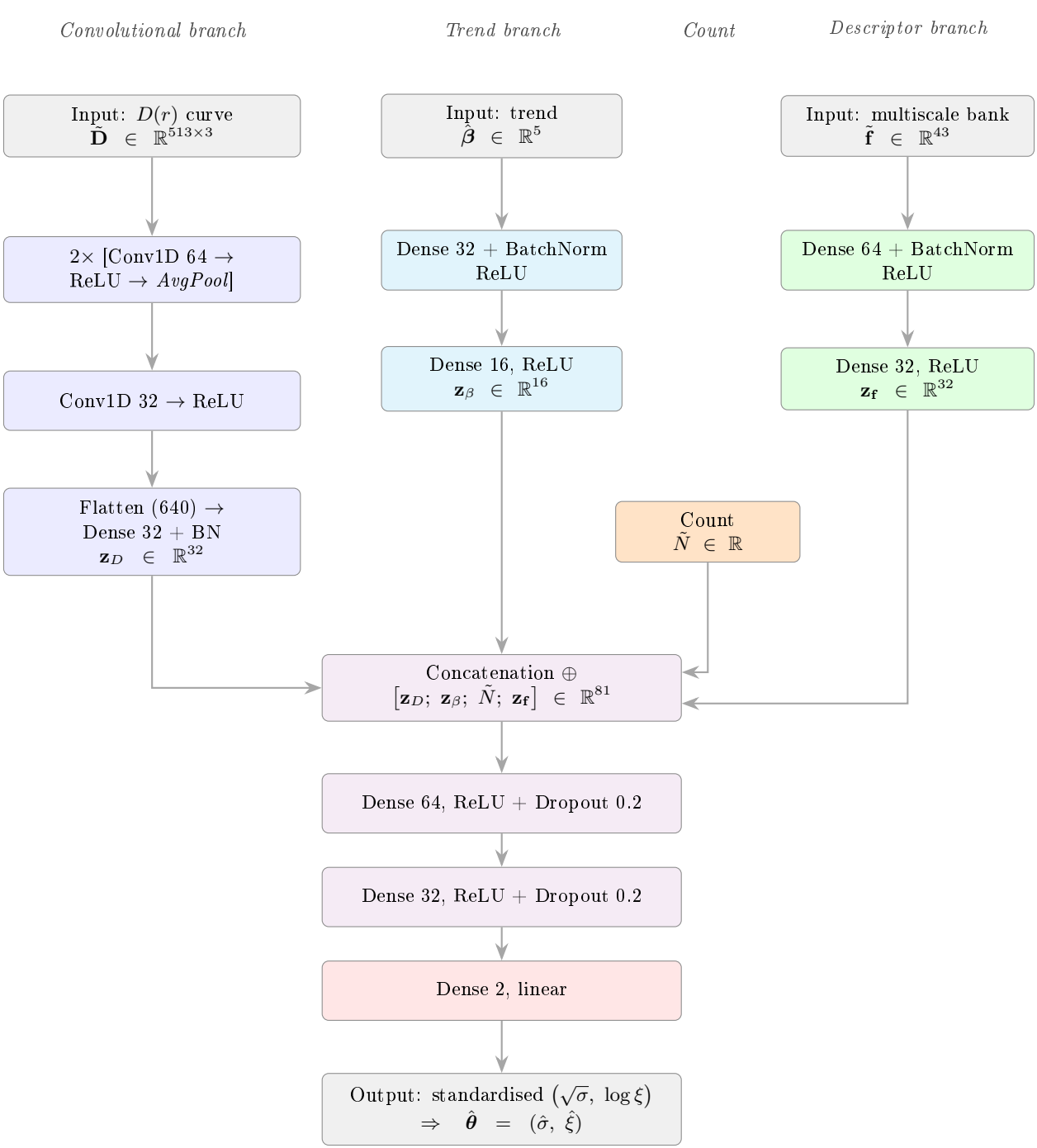

Convolutional branch
Trend branch
Count
Descriptor branch
Input: $D(r)$ curve
$\tilde{\mathbf{D}} \in \mathbb{R}^{513\times 3}$
Input: trend
$\hat{\boldsymbol{\beta}} \in \mathbb{R}^{5}$
Input: multiscale bank
$\tilde{\mathbf{f}} \in \mathbb{R}^{43}$
2× [Conv1D 64 → ReLU → $AvgPool$]
Dense 32 + BatchNorm ReLU
Dense 64 + BatchNorm ReLU
Conv1D 32 → ReLU
Dense 16, ReLU
$\mathbf{z}_\beta \in \mathbb{R}^{16}$
Dense 32, ReLU
$\mathbf{z}_\mathbf{f} \in \mathbb{R}^{32}$
Flatten (640) → Dense 32 + BN
$\mathbf{z}_D \in \mathbb{R}^{32}$
Count
$\tilde{N} \in \mathbb{R}$
Concatenation ⊕
$[\mathbf{z}_D;\ \mathbf{z}_\beta;\ \tilde{N};\ \mathbf{z}_\mathbf{f}] \in \mathbb{R}^{81}$
Dense 64, ReLU + Dropout 0.2
Dense 32, ReLU + Dropout 0.2
Dense 2, linear
Output: standardised $(\sqrt{\sigma},\ \log\xi)$
$\Rightarrow \quad \hat{\boldsymbol{\theta}} = (\hat{\sigma},\ \hat{\xi})$